

Project Plan: Capstone Project Management & Repository System (CPMRS App)

This document outlines the project plan for the **CPMRS App** (`cpmrs-app`) at **The Neotia University**.

1. Problem Statement

The Neotia University required a centralized digital system to manage **B.Tech Final Year Capstone Projects** across Cyber Security, AIML, Data Science, CSE, and ECE streams, replacing legacy manual spreadsheets and email processes.

2. Objectives

- ✓ **Digitize and automate** capstone project lifecycle workflows.
- ✓ **Dual Login System:** Support authentication via University Email or Roll Number (UID).
- ✓ **5-Digit Join Code Teaming:** Enable student team formation up to 5 members with join code lookups, join request queues, and leadership transfer.
- ✓ **Topic Selection & Proposals:** Support faculty proposal publishing and custom student project pitches.
- ✓ **Milestone Document Pipeline:** Upload & evaluation portals for synopses, interim progress reports, and final deliverables (code ZIP, report PDF, presentation slides).
- ✓ **Security & Auditability:** Next.js Edge Middleware route guards, HTTP-only JWT cookies, and immutable activity audit logs (`AuditLog`).

3. Technology & Technical Scope (`cpmrs-app`)

- **Next.js 16 App Router** with React 19, TypeScript 5, Tailwind CSS 4, and Lucide Icons.
- **Prisma 6 ORM** with **16 Relational Models** mapped to PostgreSQL.

- **11 API Route Module Directories** in `src/app/api/` (`auth` , `student` , `faculty` , `admin` , `coordinator` , `hod` , `super-admin` , `profile` , `templates` , `upload` , `landing-stats`).

4. Deliverables Summary

- Production web application in `cpmrs-app/` .
- Updated comprehensive documentation in `docs/` .